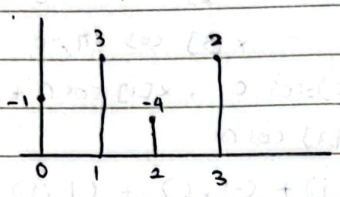
$\frac{11.28}{14} \times 100 = 80.57$

kode soal A

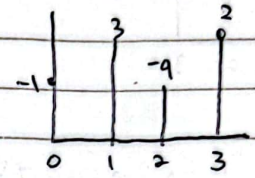
Nama : Aprilia Widuri P
Nim : 2213 4 4004.
Kelas : 2A - TNK 1.
Sinyal Sistem.

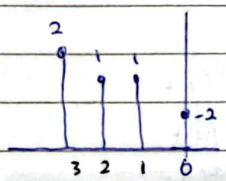
1) $x[i] = [-1\ 3\ -4\ 2]$ $\overset{0\ 1\ 2\ 3}{}$ $M = 4$
$h[k] = [-2\ 1\ 1\ 2]$. $k = 4$ $N = 4+4-1 = 7$.

konvolusi :

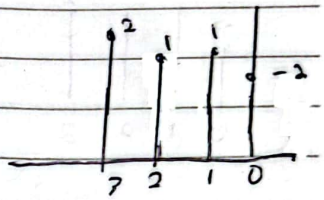


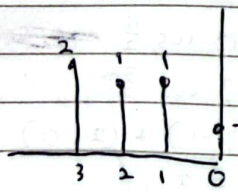

$\frac{12}{14} \times 4 = 3.43$



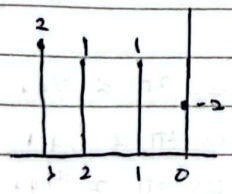

$y[0] = -2 \cdot -1 = 2$ ✓



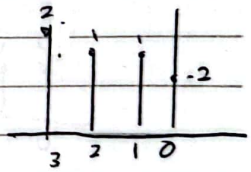

$y[1] = -1 \cdot 1 + 3 \cdot -2$
$= -1 - 6 = -7$ ✓

$y[5] = 2 \cdot 1 + -4 \cdot 2$
$= 2 - 8$
$= -6$ ✓



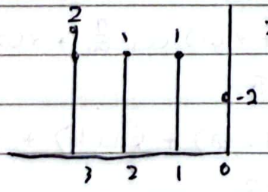

$y[2] = 4 \cdot (-2) + (3 \cdot 1) + (-1 \cdot 1)$
$= -8 + 3 - 1$
$= -9 + 3 = -6$.

=

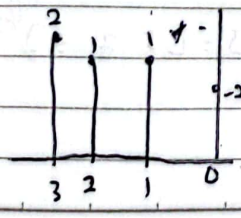



$y[6] = 2 \cdot 2 = 4$ ✓

$y[n] = [2\ -7\ -6\ -11\ 4\ -6\ 4]$




$y[3] = (2 \cdot -2) + (-4 \cdot 1) + (3 \cdot 1)$
$+ (1 \cdot 2)$
$= -4 - 4 + 3 + 2$
$= -16 + 5$
$= -11$.




$y[4] = (2 \cdot 1) + (-4 \cdot 1) + 3 \cdot 2$
$= 2 + (-4) + 6$
$= 4$ ✓

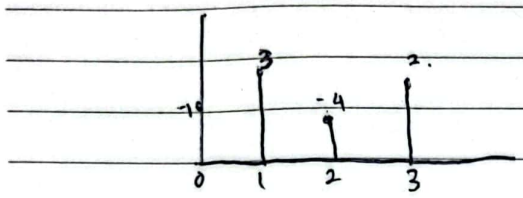
SIDU

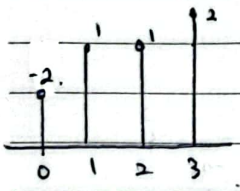
4

korelasi

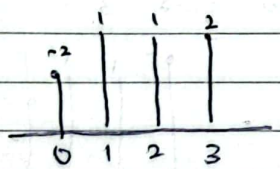
l = [-1 3 -4 2]  M = 7.

k = [-2 1 1 2].

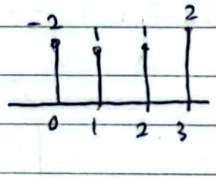


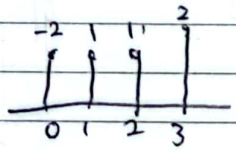

y[0] = -1 . 2 = -2 ✓



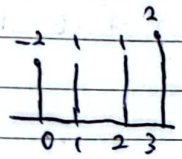

y[1] = 3 . 2 + (-1) . 1
= 6 - 1 = 5 ✓



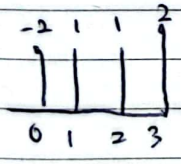

y[2] = -4 . 2 + 3 . 1 + (-1) . 1
= -8 + 3 - 1
= -9 + 3 = -6. ✓



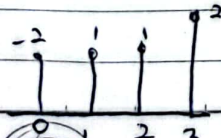

y[3] = 2 . 2 + -4 . 1 + 3 . 1
+ -1 . -2
= 4 - 4 + 3 + 2
= 5 ✓




y[4] = 2 . 1 + (-4) . 1 +
3 . -2
= 2 - 4 - 6
= -8. ✓




y[5] = 2 . 1 + -4 . -2
= 2 + 8
= 10 ✓




y[6] = 2 . -2
= -4 ✓

y[n] = [-2 5 -6 5 -8 10]

[3] fungsi dalam domain waktu.

x[n] = [3 -1 1 2]
(0 1 2 3)

fungsi x[k]  k = 0, 1, 2.

2 domain ~~waktu~~ frekuensi

• Re x[m] dan |H| x[k]

↳ N = 4.  , k = 2.

Re x[0] = x[0] . cos 2π 0 0/4 +
x[1] cos 2π 0 1/4 +
x[2] cos 2π 0 2/4 +
x[3] cos 2π 0 3/4.
= x[0] cos 0 + x[1] cos 0 + x[2] cos 0
+ x[3] cos 0.
= (3 . 1) + (-1 . 1) + (1 . 1) + (2 . 1)
= 3 - 1 + 1 + 2 = 5 ✓

Re x[1] = x[0] cos 2π 1 . 0/4 +
x[1] cos 2π 1 . 1/4 +
x[2] cos 2π 1 . 2/4 +
x[3] cos 2π 1 . 3/4.
= x[0] . cos [0] + x[1] cos π/2 +
+ x[2] cos π + x[3] cos 3/2 π
= (3 . 1) + (-1 . 0) + (1 . -1) + (2 . 0)
= 3 + 0 + (-1) + 0 = 2. ✓

Re x[2] = x[0] cos 2π 2 . 0/4 +
x[1] cos 2π 2 1/4 +
x[2] cos 2π 2 . 2/4 +
x[3] cos 2π 2 3/4.
= x[0] cos 0 + x[1] cos π + x[2] cos 2π
+ x[3] . cos 3π.
= (3 . 1) + (-1 . -1) + (1 . 1) + (2 . -1)
= 3 + 1 + 1 - 2
= 5 - 2 = 3 ✓

∴ Re x[k] = [5 2 3] h

Imaginer. $x[n] = \overset{0}{[}\ 3\ \overset{1}{-1}\ \overset{2}{1}\ \overset{3}{2}\ ]$

$\text{Im } x[0] = -[x[0] \sin 2\pi \cdot 0 \cdot 0/4 +$
$x[1] \sin 2\pi \cdot 0 \cdot 1/4 +$
$x[2] \sin 2\pi \cdot 0 \cdot 2/4 +$
$x[3] \cdot \sin 2\pi \cdot 0\ 3/4]$

$= -[x[0] \sin 0 + x[1] \sin 0 + x[2] \sin 0 + x[3] \sin 0]$

$= 3.0 + -1.0 + 1.0 + 2.0 = 0.$ ✓

$\text{Im } x[1] = -[x[0] \sin 2\pi \cdot 1 \cdot 0/4 +$
$x[1] \sin 2\pi \cdot 1 \cdot 1/4 +$
$x[2] \sin 2\pi \cdot 1 \cdot 2/4 +$
$x[3] \sin 2\pi \cdot 1 \cdot 3/4]$

$= -[x[0] \sin 0 + x[1] \sin \frac{\pi}{2} + x[2] \sin \pi + x[3] \sin \frac{3}{2}\pi]$

$= -[(3 \cdot 0) + (-1 \cdot 1) + (1 \cdot 0) + (2 \cdot -1)]$

$= -[0 + (-1) + 0 + (-2)]$

$= -[-3] = 3$ ✓

$\text{Im } x[2] = -[x[0] \sin 2\pi \cdot 2 \cdot 0/4 +$
$x[1] \sin 2\pi \cdot 2 \cdot 1/4 +$
$x[2] \sin 2\pi \cdot 2 \cdot 2/4 +$
$x[3] \sin 2\pi \cdot 2 \cdot 3/4]$

$= -[x[0] \sin 0 + x[1] \sin \pi + x[2] \sin 2\pi + x[3] \sin 3\pi]$

$= -[(3 \cdot 0) + (-1 \cdot 0) + (1 \cdot 0) + (2 \cdot 0)]$

$= 0$

$\therefore \text{Im } x[k] = [0\ 3\ 0]$ ✓

$\frac{9}{10} \times 4 = 3.6$

④. Re X[k] : [4 -2 1] (k = 0 1 2) N = 4.

Im x [k] = [0 -2 0]. x[i] = 0, 1, 2, 3

Jawab:

| | |
|---|---|
| $Re\,\bar{x}[0] = \frac{Re\,x[0]}{N} = \frac{4}{4} = 1$ ✓ | $Im\,\bar{x}[0] = -\frac{Im\,x[0]}{N/2} = -\frac{0}{4/2} = 0$ ✓ |
| $Re\,\bar{x}[1] = \frac{Re\,x[1]}{N/2} = \frac{-2}{4/2} = -1.$ ✓ | $Im\,\bar{x}[1] = -\frac{Im\,x[1]}{N/2} = -\frac{-2}{4/2} = \frac{2}{2} = 1$ ✓ |
| $Re\,\bar{x}[2]$ (↓ N/2.) $= \frac{Re\,x[2]}{N} = \frac{1}{4} = 0.25$ ✓ | $Im\,\bar{x}[2] = -\frac{Im\,x[2]}{N/2} = -\frac{0}{4/2} = 0.$ ✓ |

$x[0] = Re\,\bar{x}[0] \cdot \cos(2\pi \cdot 0 \cdot 0/4) +$
$Re\,\bar{x}[1] \cdot \cos(2\pi \cdot 1 \cdot 0/4) +$
$Re\,\bar{x}[2] \cos(2\pi \cdot 2 \cdot 0/4) +$
$Im\,\bar{x}[0] \sin(2\pi \cdot 0 \cdot 0/4 +$
$Im\,\bar{x}[1] \sin(2\pi \cdot 1 \cdot 0/4) +$
$Im\,\bar{x}[2] \sin(2\pi \cdot 2 \cdot 0/4)$

$= 1 \cdot \cos 0 + (-1) \cdot \cos 0 + 0.25 \cos 0 + 0 \cdot \sin 0 + 1 \sin 0 + 0 \sin 0$

$= 1 \cdot 1 + ((-1) \cdot 1) + (0.25 \cdot 1) + 0 + 0 + 0$

$= 1 - 1 + 0.25 + 0 + 0 + 0$

$= 0.25$ ✓

$x[1] = Re\,\bar{x}[0] \cdot \cos(2\pi \cdot 0 \cdot 1/4)$
$Re\,\bar{x}[1] \cdot \cos(2\pi \cdot 1 \cdot 1/4)$
$Re\,\bar{x}[2] \cdot \cos(2\pi \cdot 2 \cdot 1/4)$
$Im\,\bar{x}[0] \cdot \sin(2\pi \cdot 0 \cdot 1/4)$
$Im\,\bar{x}[1] \sin(2\pi \cdot 1 \cdot 1/4)$
$Im\,\bar{x}[2] \sin(2\pi \cdot 2 \cdot 1/4)$

$= 1 \cos 0 + (-1) \cos \frac{\pi}{2} + 0.25 \cos \pi + 0 \sin 0 + 1 \sin \frac{\pi}{2} + 0 \sin \pi$

$= 1 \cdot 1 + (-1) \cdot (0) + (0.25 \cdot -1) + 0 + (1 \cdot 1) + 0$

$= 1 + 0 - 0.25 + 0 + 1 + 0$

$= 2 - 0.25 = 1.75$ ✓

$x[2] = Re\,\bar{x}[0] \cdot \cos(2\pi \cdot 0 \cdot 2/4) +$
$Re\,\bar{x}[1] \cdot \cos(2\pi \cdot 1 \cdot 2/4) +$
$Re\,\bar{x}[2] \cdot \cos(2\pi \cdot 2 \cdot 2/4) +$
$Im\,\bar{x}[0] \cdot \sin(2\pi \cdot 0 \cdot 2/4) +$
$Im\,\bar{x}[1] \cdot \sin(2\pi \cdot 1 \cdot 2/4) +$
$Im\,\bar{x}[2] \sin(2\pi \cdot 2 \cdot 2/4)$

$= 1 \cos 0 + -1 \cos 2\pi(1/2 +$
$0.25 \cos 2\pi + 0 \sin 0 + 1 \sin \frac{2\pi}{2}$
$+ 0 \sin 2\pi$
$= (1 \cdot 1) + (-1) \cdot (-1) + (0.25 \cdot 1) +$
$(0 \cdot 0) + (0 \cdot 0) + 0 \cdot 0$
$= 1 + 1 + 0.25 + 0 + 0 + 0$
$= 2.25$ ✓

SIDU

Apnira Widuri P

2213 440041

2A . TUK.

Kode A

$x[3] = Re\ \bar{x}[0] \cos 2\pi\ 0\ 3/4 +$

$Re\ \bar{x}[1] \cos 2\pi\ 1\ 3/4 +$

$Re\ \bar{x}[2] \cos 2\pi\ 2\ \frac{3}{4} +$

$Im\ \bar{x}[0] \cos 2\pi\ 0\ 3/4 +$

$Im\ \bar{x}[1] \cos 2\pi\ 1\ 3/4 +$

$Im\ \bar{x}[2] \cos 2\pi\ 2\ 3/4$

$= 1 \cos 0 + (-1) \cos \frac{3}{2}\pi + 0{,}125 \cos 3\pi + 0 \sin 0 + 1 \sin \frac{3}{2}\pi + 0 \sin 3\pi$

$= 1.1 + (-1)(0) + 0{,}125(-1) + 0 + (1.-1) + 0$

$= 1 + (-1) - 0{,}125 + 0 + (-1) + 0$

$= -1{,}125$ ✓

$x[n] = [0{,}125 \quad 1{,}75 \quad 2{,}125 \quad -1{,}125]$. ✓ ✓

2.).

| | | |
|---|---|---|
| 1 | -2 | 3 |
| -4 | 5 | -6 |

a:

| | |
|---|---|
| 2 | 1 |
| -1 | 2 |
| 2 | 1 |

→

| | |
|---|---|
| 1 | 2 |
| 2 | -1 |
| 1 | 2 |

→ x

| | |
|---|---|
| 1 | 2 |
| 2 | -1 |
| 1 | 2 |

↓ y

→ x

$M = x + I - 1$

$= 2 + 3 - 1$

$= 4$ kolom

$N = 3 + 2 - 1$

$= 4$ baris

$\frac{1}{16} \times 4 = 0.25$

| | |
|---|---|
| 1 | 2 |
| 2 | -1 |
| 1 | 2 |

| | | |
|---|---|---|
| -1 | -2 | 3 |
| -4 | 5 | -6 |

$y(0,0) = 2.1 = 2$. ✓

$y(1,0) = 2.-2 + 2.1 = -4 + 2 = -2$.